

Engineering Notes

Robust Stall Spin Flight Path Control with Flight Test Validation

Jeremy W. Hopwood,*[✉] James L. Gresham,[†]
and Craig A. Woolsey[‡][✉]
Virginia Tech, Blacksburg, Virginia 24061

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I. Introduction

IT IS often required that a fixed-wing unmanned aircraft, such as the RQ-4 Global Hawk, be equipped with an emergency flight termination sequence in the case of a malfunction, such as loss of propulsion or communication [1]. One common flight termination approach is to put the aircraft in a stall spin—a steady descent toward a small and predictable impact zone and with a relatively low impact speed as compared to a gliding or spiral descent. Such a termination approach has been incorporated into the flight control system of NASA's Airborne Subscale Transport Aircraft Research (AirSTAR) platform for beyond visual line-of-sight research [2], and has been a requirement in the AIAA Student Design, Build, Fly Competition [3]. The main advantage of a spin is that it dissipates energy in a stable, predictable, and controlled manner and thus is an effective failsafe [4].

Still, it is desirable to have the ability to control and direct this spinning descent along a prescribed path toward an acceptable impact zone as suggested in [5]. This paper presents the development, simulation, and flight test of a robust, linear control law to direct the spinning descent of an aircraft along a periodic inertial path with a desired average azimuth and inclination angle. The aircraft is controlled by two feedback control laws running in parallel, each with a specialized objective.

First, an infinite-horizon yaw-periodic linear quadratic (LQ) controller computes the optimal input that minimizes a multi-objective cost of cumulative velocity vector error and input/state perturbation from a steady vertical spin. Working in isolation, this approach effectively slants the otherwise helical descent path so that the overall motion is toward a preferred azimuthal direction. However, this control law tends to command inputs beyond the physical limits of the system when exposed to external disturbances and modeling errors.

Second, a robust H_∞ controller places an upper bound on the worst-case map from exogenous forces, moments, and angular deflections of control surfaces produced by the LQ controller to the aircraft's perturbation from the nominal spin (or rather a frequency-

dependent weighting of this). The model is assumed to have time-invariant uncertainty in the aerodynamic parameters, representing very real uncertainty and unpredictability of spin aerodynamics [6]. D-K iteration is used to synthesize the robust H_∞ controller, yielding a dynamic output-feedback control law [7].

The driving factor in choosing this control architecture is the need for good tracking of a desired skewed helical descent while also robustly maintaining a spin. Too much emphasis on tracking in the control design tends to drive the aircraft out of a spin into a dive, while a high penalty on perturbation from the vertical spin results in poor tracking. Additionally, because the aerodynamics in a spin are generally unknown, the control law must be robust to such model perturbations.

The remainder of this paper is organized as follows. Section II presents the mathematical model of the aircraft used in this paper. Section III details the design of the proposed parallel control law. Section IV presents both simulation and flight test validation of the control law demonstrating robust tracking of a linear path in a spin, and Sec. V gives concluding remarks.

II. Aircraft Dynamic Model

The aircraft used in the design, simulation, and flight test of this controller was an E-Flite Carbon-Z Cessna 150 (CZ150) radio-controlled aircraft, instrumented as described in [8,9], as shown in Fig. 1. This paper uses the flight-derived nominal glide model for the CZ150 detailed in [8]. The CZ150 mass and geometric properties, tabulated in Table 1, were determined with methods described in Ref. [10].

The aircraft is modeled as a rigid body of mass m with three control surfaces (ailerons, elevators, rudder). It is assumed the throttle is set to zero during the flight termination maneuver. Let unit vectors $\{\hat{i}_1, \hat{i}_2, \hat{i}_3\}$ define an Earth-fixed north-east-down (NED) orthonormal reference frame $\mathcal{F}_1$. Let the unit vectors $\{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$ define the orthonormal body-fixed frame $\mathcal{F}_B$ centered at the aircraft center of mass with $\hat{b}_1$ pointing along the longitudinal axis out the nose of the aircraft, $\hat{b}_2$ out the right wing, and $\hat{b}_3$ out the bottom of the aircraft. The position of the body frame with respect to the inertial frame is given by the vector $\mathbf{r} = [x \ y \ z]^T$. The attitude of the aircraft is given by the rotation matrix $\mathbf{R}_{IB}$ that maps free vectors in $\mathcal{F}_B$ to $\mathcal{F}_1$. Consider the 1-2-3 rotation

$$\mathbf{R}_{IB} = e^{\hat{e}_3\psi} e^{\hat{e}_2\theta} e^{\hat{e}_1\phi} \quad (1)$$

parameterized by roll (ϕ), pitch (θ), and yaw (ψ) angles, where $\mathbf{e}_1 = [1 \ 0 \ 0]^T$, etc., and $\hat{\cdot}$ is the skew-symmetric cross-product equivalent matrix satisfying $\hat{\mathbf{a}}\mathbf{b} = \mathbf{a} \times \mathbf{b}$. Let $\mathbf{v} = [u \ v \ w]^T$ and $\boldsymbol{\omega} = [p \ q \ r]^T$ be the translational and rotational velocity of the aircraft with respect to $\mathcal{F}_1$ expressed in $\mathcal{F}_B$, respectively. With this choice of Euler angles, we have the kinematic equations of motion:

$$\dot{\mathbf{r}} = \mathbf{R}_{IB}\mathbf{v} \quad (2a)$$

$$\dot{\mathbf{R}}_{IB} = \mathbf{R}_{IB}\hat{\boldsymbol{\omega}} \quad (2b)$$

With $\boldsymbol{\Theta} := [\phi \ \theta \ \psi]^T$, Eq. (2b) becomes

$$\dot{\boldsymbol{\Theta}} = \underbrace{\begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}}_{\mathbf{L}_{IB}} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (3)$$

Let us represent the aerodynamic forces and moments on the aircraft expressed in $\mathcal{F}_B$ as $\mathbf{F} = [X \ Y \ Z]^T$ and $\mathbf{M} = [\mathcal{L} \ \mathcal{M} \ \mathcal{N}]^T$.

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*Ph.D. Candidate, Kevin T. Crofton Department of Aerospace and Ocean Engineering; jeremyhopwood@vt.edu. Student Member AIAA.

[†]Postdoctoral Research Affiliate, Kevin T. Crofton Department of Aerospace and Ocean Engineering; also Flight Test Engineer and Lieutenant Colonel, United States Air Force; jamesgresham@vt.edu. Senior Member AIAA.

[‡]Professor, Kevin T. Crofton Department of Aerospace and Ocean Engineering; cwoolsey@vt.edu. Associate Fellow AIAA.



Fig. 1 E-Flite Carbon-Z Cessna 150.

Table 1 CZ150 mass and geometric properties

Property	Symbol	Value	Units
Mass	m	0.336	slug
Mean aerodynamic chord	$\bar{c}$	1.049	ft
Projected wing span	b	6.97	ft
Projected wing area	S	7.31	ft ²
Roll moment of inertia	I_{xx}	0.457	slug-ft ²
Pitch moment of inertia	I_{yy}	0.357	slug-ft ²
Yaw moment of inertia	I_{zz}	0.622	slug-ft ²
Product of inertia	I_{xz}	0.052	slug-ft ²
Products of inertia	I_{xy}, I_{yz}	~ 0	slug-ft ²

Define $\mathbf{p} = m\mathbf{v}$ to be the linear momentum of the aircraft and $\mathbf{h} = \mathbf{I}\boldsymbol{\omega}$ to be the angular momentum vector about the center of mass, both expressed in $\mathcal{F}_B$. Here, $\mathbf{I}$ is the moment of inertia matrix about the center of mass in $\mathcal{F}_B$. Thus in a mixed momentum/velocity notation, we have the dynamic equations of motion:

$$\dot{\mathbf{p}} = \mathbf{p} \times \boldsymbol{\omega} + m\mathbf{g}\mathbf{R}_{BI}\mathbf{e}_3 + \mathbf{F} \quad (4a)$$

$$\dot{\mathbf{h}} = \mathbf{h} \times \boldsymbol{\omega} + \mathbf{M} \quad (4b)$$

Requiring a high-fidelity spin aerodynamic model in order to design the proposed model-based flight termination strategy would be unrealistic for most aircraft. For this reason, we use a nonlinear aerodynamic model validated for large perturbations from nominal, wings-level, gliding flight [8]. We assume that this model sufficiently captures the dominant dynamics—a claim that is later investigated through flight tests. The body-frame forces and moments are functions of airspeed, $V = \|\mathbf{v}\|$; air density, ρ ; angle of attack, α ; angle of sideslip, β ; body angular velocity, $\boldsymbol{\omega}$; and control surface deflections, $\boldsymbol{\delta} = [\delta_a \ \delta_e \ \delta_r]^T$. Here, α and β parameterize the transformation from the wind frame $\mathcal{F}_W$ to the body frame, given by the rotation matrix

$$\mathbf{R}_{BW} = e^{-\hat{\mathbf{e}}_2\alpha}e^{\hat{\mathbf{e}}_3\beta} \quad (5)$$

A nominal stall spin is a steady motion for which yaw angle increases monotonically in time. We define the *reduced state* vector $\mathbf{x}_r$ to be the states that are constant in a spin: roll and pitch angle, body velocity, and body angular velocity. We desire the nominal spin to be a steep stall spin with aileron, elevator, and rudder at about 70% maximum deflection, which was chosen using a combination of the methods presented in [11] and from flight data. These requirements correspond to control surface deflections that are not too close to the saturation limits, but also result in a relatively low angle of attack to maintain aerodynamic control authority and model validity. With an aerodynamic model, the nominal spin can readily be found using the model-based methods detailed in [11,12]. However, for the aircraft presented here, the nominal spin was chosen from flight data collected in conjunction with the aerodynamic system identification flight tests presented in [8]. Steady nominal spin state and control input values for the CZ150 determined from flight data are

$$\begin{aligned} \tilde{\phi} &= -24.01 \text{ deg} & \tilde{\theta} &= -73.96 \text{ deg} \\ \tilde{V} &= 66.93 \text{ ft/s} & \tilde{\alpha} &= 12.38 \text{ deg} & \tilde{\beta} &= 5.74 \text{ deg} \\ \tilde{p} &= -384.5 \text{ deg/s} & \tilde{q} &= 36.23 \text{ deg/s} & \tilde{r} &= -97.17 \text{ deg/s} \\ \tilde{\delta}_a &= 15.5 \text{ deg} & \tilde{\delta}_e &= -18.5 \text{ deg} & \tilde{\delta}_r &= 24.6 \text{ deg} \end{aligned} \quad (6)$$

where $\tilde{\cdot}$ denotes the nominal quantity. Note that the nominal values given in Eq. (6) do not correspond precisely to an equilibrium of the reduced system, but will be very close. The consequences of this observation are further discussed at the end of Sec. III.C.

For the nominal spin chosen, yaw angle takes the form

$$\tilde{\psi}(t) = \psi_0 + \tilde{\Omega}t \quad (7)$$

where, from Eq. (3), we have the nominal spin rate

$$\tilde{\Omega} := \tilde{\dot{\psi}} = \tilde{q} \sin \tilde{\phi} \sec \tilde{\theta} + \tilde{r} \cos \tilde{\phi} \sec \tilde{\theta} \approx -375 \text{ deg/s} \quad (8)$$

Then, from Eq. (2a), the vertical descent rate is steady with

$$\tilde{z} = -\tilde{u} \sin \tilde{\theta} + \tilde{v} \sin \tilde{\phi} \cos \tilde{\theta} + \tilde{w} \cos \tilde{\phi} \cos \tilde{\theta} \approx 64.3 \text{ ft/s} \quad (9)$$

In a nominal stall spin, the lateral displacement dynamics from Eq. (2a) reduce to

$$\dot{\tilde{x}} = -C_1 \sin \psi + C_2 \cos \psi \quad (10a)$$

$$\dot{\tilde{y}} = C_2 \sin \psi + C_1 \cos \psi \quad (10b)$$

where C_1 and C_2 are constant for the nominal spin. Thus for the nominal motion, we have

$$\tilde{x}(t) = x_0 + \frac{1}{\tilde{\Omega}}(C_1(\cos \psi - \cos \psi_0) + C_2(\sin \psi - \sin \psi_0)) \quad (11a)$$

$$\tilde{y}(t) = y_0 + \frac{1}{\tilde{\Omega}}(-C_2(\cos \psi - \cos \psi_0) + C_1(\sin \psi - \sin \psi_0)) \quad (11b)$$

where x_0 , y_0 , and ψ_0 are the initial values of x , y , and ψ , respectively.

III. Controller Design

A. Approach

We choose to define the desired path to be a constant unit vector, defined from the initiation of the developed spin, pointing in some desired direction. The aim is to slant the centerline of this helical descent to converge on the specified line. The desired path is minimally represented by an azimuth angle φ_d and inclination angle θ_d . As a vector in the inertial frame $\mathcal{F}_I$, this direction is denoted $\boldsymbol{\sigma}_d$. The goal of the controller is to make the perpendicular distance between the aircraft center of mass and the desired path small, while ensuring that the aircraft state variable values remain sufficiently close to values that correspond to the nominal spin. Some major challenges are as follows:

- i) An accurate aerodynamic model for a spin is generally not readily known.
- ii) Control surface deflections are easily saturated in a spin.
- iii) For a skewed helical spin about the desired path, the cross-track error of the aircraft is not identically zero.
- iv) A linear time-varying approach may not be appropriate in a stall spin because small perturbations in yaw rate cause yaw angle to drift from nominal, leading to inaccuracy of the small perturbation model over time.

The dynamics are augmented with the integral of the error between the desired and actual velocity vector directions to address (iii). Through an independent variable change from time to yaw angle, a periodic LQ regulator is designed to address (iv). Then, a linear time-invariant H_∞ controller is designed for the reduced dynamics to address (i) and (ii). Therefore, we propose the control

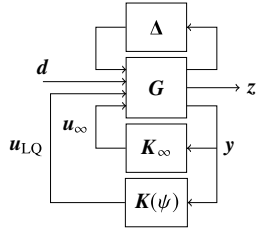


Fig. 2 Parallel control architecture.

architecture shown in Fig. 2, where Δ is an operator that belongs to a chosen uncertainty set $\mathbb{A}$, interconnected with the performance-based, linear aircraft dynamics model G . Here, K_∞ is the dynamic H_∞ controller, $K(\psi)$ is the yaw-periodic LQ state feedback gain, y are the measurements, z are the performance outputs that we want to make small, and d are the exogenous force and moment disturbances to the system.

B. Yaw-Periodic Linear Quadratic Regulator

To address the periodic nature of the cross-track error dynamics, we choose to augment the system with a state that represents the integral of the error between the desired and actual velocity vector directions. The velocity vector direction expressed in $\mathcal{F}_1$ is

$$\sigma_i = R_{1W} e_1 \quad (12)$$

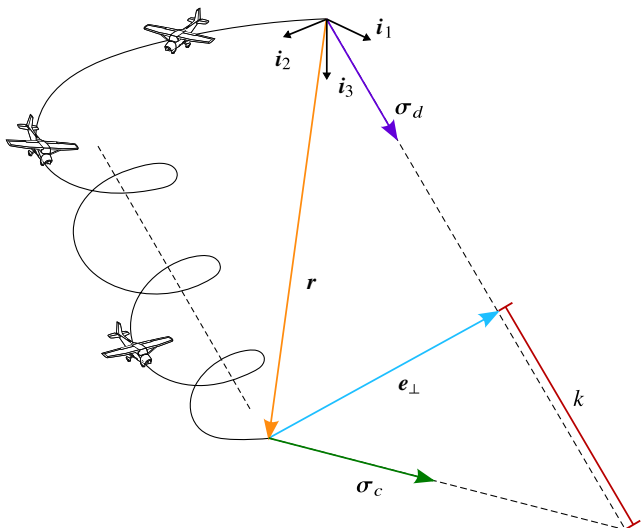
where $R_{1W} = R_{1B} R_{BW}$. Because the desired path is linear, the cross-track error can be defined using the projection of the position vector onto the desired direction, σ_d , yielding

$$e_\perp = (r^\top \sigma_d) \sigma_d - r \quad (13)$$

The position vector r is defined with respect to $\mathcal{F}_1$ fixed at the initiation of the control law as shown in Fig. 3. The corrected velocity vector direction σ_c is then defined to be the unit vector that points from the aircraft to a location on the desired path some look-ahead distance k ahead of the intersection of the lines defined by σ_d and $e_\perp$. This is illustrated in Fig. 3 and expressed as

$$\sigma_c = \frac{e_\perp + k\sigma_d}{\|e_\perp + k\sigma_d\|} \quad (14)$$

Pointing the desired velocity vector direction toward the path ensures that the spin axis of the convergent trajectory is on the path as opposed

Fig. 3 Cross-track error $e_\perp$ and corrected desired velocity vector direction σ_c .

to parallel to it, which would be the case if $\sigma_c \equiv \sigma_d$. Then, defining e_v to be the integral of the error between the actual and corrected velocity vector directions, we have

$$\dot{e}_v = \sigma_i - \sigma_c \quad (15)$$

where σ_i is the actual velocity vector direction. We now define the augmented equations of motion from Eqs. (3), (4), and (15) as

$$\dot{\Theta} = L_{1B} \omega \quad (16a)$$

$$\dot{v} = v \times \omega + g R_{1B} e_3 + \frac{1}{m} F \quad (16b)$$

$$\dot{\omega} = I^{-1} (I \omega \times \omega + M) \quad (16c)$$

$$\dot{e}_v = \sigma_i - \sigma_c \quad (16d)$$

Linearizing these dynamics about the nominal spinning motion gives a T -periodic linear state equation, where $T = |360/\tilde{\Omega}| \approx 0.9$ s. Since ψ is a fast-varying state variable for this motion, the difference between the true heading $\psi(t)$ and the nominal value $\tilde{\psi}(t)$ can quickly grow large. This results in poor azimuth tracking by a time-periodic controller. Consider the dynamics in Eq. (16), where the position r , appearing only in σ_c , is assumed to be nominal as a function of yaw angle given by $r = \tilde{r}(\psi)$ from Eq. (11). Equations (16a–16c) are time invariant, and $\tilde{\psi}(t)$ is the only time-varying nominal state variable that explicitly appears in the augmented dynamics through Eq. (16d). Because $\psi(t)$ increases linearly in time during a nominal stall spin, we may consider the independent variable change $\psi = \tilde{\psi}(t)$. This gives a system in the form

$$x'(\psi) = \frac{1}{\tilde{\Omega}} \begin{bmatrix} f_{x,r}(x_r, u_{LQ}) \\ f_{e,v}(\tilde{r}(\psi), x_r, \psi) \end{bmatrix} \quad (17)$$

where $x := [x_r^\top \ e_v^\top]^\top$ and $(\cdot)' = d(\cdot)/d\psi$. When $\psi = \tilde{\psi}(t) = \tilde{\Omega}t$, this system produces trajectories that are equivalent to those of the original system, but with the independent variable scaled by the nominal spin rate. From Eq. (17), we have the 2π -periodic linearization in ψ :

$$\delta x'(\psi) = A_a(\psi) \delta x(\psi) + B_a \delta u_{LQ}(\psi) \quad (18)$$

where δ denotes a perturbation from the nominal value. Equation (18) is the linear state equation for which the ψ -varying LQ controller is designed. The linear-quadratic regulator problem is to determine the control u_{LQ} that minimizes

$$J = \lim_{\psi_f \rightarrow \infty} \int_0^{\psi_f} (\delta u_{LQ}^\top(\psi) R \delta u_{LQ}(\psi) + \delta x^\top(\psi) Q \delta x(\psi)) d\psi \quad (19)$$

subject to Eq. (18) and initial condition $\delta x(0) = \delta x_0$. Consider the choice of

$$Q = \begin{bmatrix} \text{diag}(\delta x_{r,\max})^{-1} & \mathbf{0} \\ \mathbf{0} & \nu \text{diag}(\tilde{\sigma}) \end{bmatrix} \quad R = \rho \text{diag}(\delta u_{\max})^{-1}$$

where $\tilde{\sigma}$ is a nominal unit magnitude direction vector to normalize the penalty on horizontal and vertical velocity error. Here, $\delta(\cdot)_{\max}$ represents the desired perturbation limits by which the cost function is normalized. The tuning parameters ν and ρ help balance the three objectives: reduced state perturbation, input perturbation, and cross-track error. Let $P(\psi, \psi_f)$ be the solution of the differential Riccati equation: $-P' = A_a^\top P + P A_a - P B_a R^{-1} B_a^\top P + Q$, with $P(\psi_f) = \mathbf{0}$. Since $A_a(\psi)$ is bounded and continuous and the pair (A_a, B_a) is controllable, $P(\psi, \psi_f)$ converges to some bounded, positive semidefinite $\bar{P}(\psi)$ as $\psi_f \rightarrow \infty$ [13]. Then the solution to the LQ control design problem is

$$\delta u_{LQ}(\psi) = -\underbrace{R^{-1} B_a^\top \tilde{P}(\psi)}_{K(\psi)} \delta x(\psi) \quad (20)$$

Values for the three control parameters ν , ρ , and k are chosen through simulation of the LQ controller alone (without disturbances and model perturbations) such that inputs do not saturate and there is a good balance between reduced state perturbation from the nominal spin and cross-track error. For the CZ150 used in the simulation and flight test of this control law, we chose $\nu = 1$, $\rho = 1$, and $k = 350$ ft. For this system, the numerically approximated steady-state solution of the Riccati equation yielded a feedback gain $K(\psi)$, where each entry is either constant or sinusoidal in ψ with a period of 2π .

C. Robust H_∞ Controller

A robust H_∞ controller is now designed for the linear, time-invariant reduced state dynamics. Consider the reduced linear system subject to the following three inputs: LQ controller commanded inputs, u_{LQ} ; force and moment disturbances, d ; and the remaining control inputs to be assigned according to the H_∞ control law, u_∞ . The force and moment disturbances are assumed to be finite-time square-integrable ($d(t) \in \mathbb{R}^6$, $d \in L_{2,e}^6$), and the commanded input from the LQ controller, u_{LQ} , is an exogenous, measurable signal. This system, denoted P , is defined by the linear state equation

$$P: \delta \dot{x}_r = A_r \delta x_r + B_r (\delta u_{LQ} + \delta u_\infty) + B_d d \quad (21)$$

with measurements $y = [\delta x_r^\top \ \delta u_{LQ}^\top]^\top$. We define the normalized exogenous disturbances w and performance outputs z , respectively, in the Laplace domain as

$$\begin{aligned} w(s) &= \begin{bmatrix} W_d^{-1}(s) & 0 \\ 0 & W_{LQ}^{-1}(s) \end{bmatrix} \begin{bmatrix} d(s) \\ \delta u_{LQ}(s) \end{bmatrix} \\ z(s) &= \begin{bmatrix} W_x(s) & 0 \\ 0 & W_u(s) \end{bmatrix} \begin{bmatrix} \delta x_r(s) \\ \delta u_\infty(s) \end{bmatrix} \end{aligned} \quad (22)$$

where W_d , W_{LQ} , W_x , and W_u are frequency-dependent scalings based on maximum allowable/expected values of the corresponding variables. For the proceeding control law synthesis, the maximum allowable reduced state perturbation and maximum expected forces and moments are taken from flight data of a control-perturbed spin [8].

Additionally, W_d and W_{LQ} are scaled by a weighting factor, $\lambda \in (0, 1)$, which represents the weighting of force/moment disturbance rejection relative to the LQ controller attenuation. This weighting, as shown in Fig. 4, gives a realistic performance criterion to be robustly satisfied. As λ approaches 0, the closed-loop system only has exogenous inputs from the LQ controller. Conversely, as λ approaches 1, the system only has exogenous disturbance inputs. For simulation, we choose a disturbance weighting of $\lambda = 0.5$ to equally weight force/moment disturbance rejection and worst-case LQ controller attenuation.

Since the aerodynamics are uncertain, the control law is designed to be robust to the uncertainty of aerodynamic model parameters. We assume that the uncertainty acts on the plant G as a linear bounded operator Δ in a way that represents the time-invariant uncertainty in

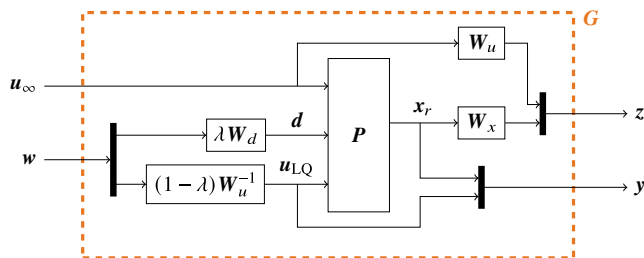


Fig. 4 Performance-based system.

the 37 aerodynamic parameters found in the nominal glide model in [8]. Therefore, we make the following assumption:

Assumption 1: The true, unknown dynamics linearized about the nominal spin is contained within the subspace of linear systems defined by the uncertain representation of the system, (G, Δ) .

Let M be the closed-loop, transfer function from (p, w) to (q, z) , where $p(t) \in \mathbb{R}^p$ and $q(t) \in \mathbb{R}^p$ are the outputs and inputs of the uncertainty operator Δ , respectively. The uncertain closed-loop system is defined by the lower linear fractional transformation $M = \underline{S}(G, K_\infty) := G_{11} + G_{12}K_\infty(\mathbb{I} - G_{22}K_\infty)^{-1}G_{21}$, where

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \quad (23)$$

The interconnection with the uncertainty operator Δ is described by the upper linear fractional transformation $\bar{S}(M, \Delta) = M_{22} + M_{21}\Delta(\mathbb{I} - M_{11}\Delta)^{-1}M_{12}$, where M_{22} is the nominal closed-loop system from $w \mapsto z$. For internally stable $\delta \dot{x}_{r,cl} = M_{22}\delta x_{r,cl}$, where $\delta x_{r,cl}$ is the closed-loop reduced state perturbation, the uncertain system M is *robustly stable* when $(\mathbb{I} - M_{11}\Delta)^{-1}$ exists for all $\Delta \in \mathcal{A}$ [14].

When there is no uncertainty, the H_∞ problem is to find a controller K_∞ that minimizes the upper bound on the worst-case map from w to z . Specifically, we want the induced L_2 norm of the closed-loop system to satisfy

$$\|M_{22}\|_\infty = \|M_{22}\|_{L_2 \rightarrow L_2} = \text{ess sup}_{\omega \in \mathbb{R}} \bar{\sigma}(\hat{M}_{22}(j\omega)) < \gamma \leq 1 \quad (24)$$

where $\hat{M}_{22}(j\omega)$ is the frequency domain description of the nominal closed-loop system and $\bar{\sigma}(\cdot)$ is the largest singular value. Let $\mathcal{K}$ be the set of stabilizing controllers. The problem is to find

$$\arg \inf_{K_\infty \in \mathcal{K}} \|M_{22}\|_\infty < \gamma \quad (25)$$

where γ is the H_∞ performance measure. A value of γ less than 1 indicates that the closed-loop performance is contractive and our desired criterion is met.

Regarding model uncertainty, we assume that the aerodynamic parameters take on values within the 95% confidence intervals, determined from the model identification process in [8]. We choose the time-invariant uncertainty set

$$\mathcal{A} := \{\text{diag}(\Delta_1, \dots, \Delta_d) \mid \Delta_i \in \mathcal{L}_{\text{TI}}^c(L_2^{k_i}), \|\Delta_i\|_\infty < 1\} \quad (26)$$

where $\mathcal{L}_{\text{TI}}^c(L_2^k)$ is the set of causal, time-invariant, linear bounded operators on the space of k -dimensional square-integrable functions, L_2^k . Let $\mathcal{D}$ be the set of time-invariant nonzero scalings that commute with the uncertainty set $\mathcal{A}$ as shown in [14] to be

$$\begin{aligned} \mathcal{D} &= \{D \in \mathcal{L}_{\text{TI}}^c(L_2^p) \mid D \text{ non-singular,} \\ &\quad \hat{D}(s) = \text{diag}(\hat{D}_1(s)\mathbb{I}, \dots, \hat{D}_d(s)\mathbb{I})\} \end{aligned} \quad (27)$$

where each $\hat{D}_i(s) \in \text{RH}_\infty$ (the space of proper transfer functions with no poles in the closed right-half of the complex plane). With the chosen structured uncertainty, analysis and synthesis of the controller is readily conducted. Given a stabilizing controller K_∞ , we have the closed-loop uncertain system M interconnected with an operator in the uncertainty set, $\mathcal{A}$. Then, the structured singular value μ satisfies

$$\begin{aligned} \mu(M_{11}, \mathcal{A}) &= \frac{1}{\inf\{\|\Delta\|_\infty \mid \Delta \in \mathcal{A}, (\mathbb{I} - M_{11}\Delta) \text{ singular}\}} \\ &\leq \inf_{D \in \mathcal{D}} \|\hat{D} M_{11} \hat{D}^{-1}\|_\infty \end{aligned}$$

as shown in [15]. The robustly stable system has *robust performance* if $\mu(M_{11}, \mathcal{A}) < 1$, meaning that the performance measure

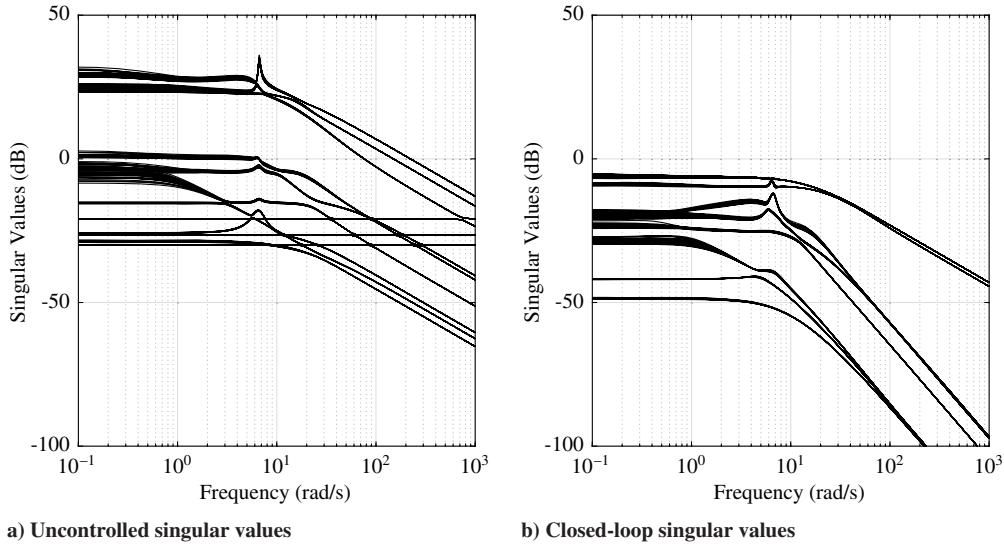


Fig. 5 Singular value comparison.

remains contractive for all structured uncertainty within the bounds determined in the modeling process. Thus, the problem is to find

$$\arg \inf_{K_\infty \in \mathcal{K}, D \in \mathcal{D}} \|\hat{D} \hat{M}_{11} \hat{D}^{-1}\|_\infty \leq \mu \quad (28)$$

This is in general a nonconvex problem, so we use D-K iteration to find a local minimum using the MATLAB® Robust Control Toolbox [16,17]. This computation yields an upper bound on the structured singular value of $\mu = 0.9066$. Therefore, the H_∞ norm of the closed-loop linear system remains below μ for all normalized structured uncertainty of $1/\mu$.

The uncontrolled and closed-loop singular value plots for a variety of uncertain parameter values are shown in Figs. 5a and 5b.

Proposition 1: The closed-loop, reduced linear system, given by the control formulations in Eqs. (20) and (28) and the connection in Fig. 2, is robustly stable.

Proof: By the μ -synthesis result of $\mu < 1$, the closed-loop reduced system is robustly stable and $\delta x_r \in L_2^8$. This holds true for exogenous signals w in extended L_2 space $L_{2,e}^9$, which satisfy $\int_0^{t_1} \|w(t)\|_2^2 dt < \infty$ for all finite $t_1 \geq 0$. The disturbance d is assumed to be in $L_{2,e}^6$, and signal u_{LQ} generated by the linear, bounded state feedback gain $K(\psi)$ is in $L_{2,e}^3$. Therefore, the closed-loop linear system is robustly stable. $\square$

Consider the closed-loop model of the nonlinear system. While the region of attraction of the feedback-stabilized spinning motion for the nonlinear closed-loop system is unknown, the robustness and performance is evaluated through simulation and validated in flight test. The benefit of this robustness is further discussed in Remark 1.

Remark 1: The point about which the reduced system is linearized is not precisely an equilibrium of the reduced nonlinear dynamics. Therefore, an internally stabilizing linear controller results in closed-loop trajectories for which the time derivative of the states is nonzero. For the nominal spinning motion (which corresponds to the origin of the linear system), this is effectively equivalent to a bounded exogenous disturbance, which is robustly rejected by the H_∞ controller.

IV. Results

Results from both simulation and flight test experiments with the control law implemented are presented here.

A. Simulation Results

The nonlinear closed-loop dynamics are simulated, incorporating control deflection bounds, wind disturbances, and perturbations from the nominal glide aerodynamic model in [8]. We illustrate the performance of this control law through the following scenario.

Suppose the Carbon-Z Cessna 150 initiates a spinning flight termination sequence with the spin fully developed at 1000 ft above ground level. There is a constant 15 knot (~ 25 ft/s) wind from the west. Through some algorithm, it is determined that the best safe crash zone is 350 ft due northeast, corresponding to approximately a 20° desired inclination angle. To demonstrate robustness, the aerodynamic parameters are arbitrarily taken as the value halfway between the estimated value and the upper 95% confidence value reported in [8]. These parameter values are used in simulation, while the control law is synthesized using the modeled parameter estimates.

Consider the closed-loop response under the yaw-periodic LQ controller, as shown in Fig. 6a, where the desired path is shown as the solid black line. The aircraft begins to track the desired path, but then exits the spin. The LQ controller cannot reject the wind disturbance and model perturbations, and it fails to keep the aircraft in the spin. To combat the lack of robustness and tendency to exit the spin, we now add the H_∞ controller in parallel. The parallel controller closed-loop response is shown in Fig. 6b, displaying good tracking of the desired path while maintaining a spin. This scenario demonstrates the benefit of the parallel control architecture. The Euclidean norm of the nondimensional performance measure for both cases, as well as constant spin inputs, is shown in Fig. 7. Both the LQ and H_∞ controllers penalize perturbations from the nominal spin, but the addition of the H_∞ control law adds robustness and a performance-based result to satisfy this requirement.

B. Flight Test Results

The control law was implemented in C++ using Robot Operating System (ROS) on a Raspberry Pi cocomputer connected to a Pixhawk flight computer running PX4 in the CZ150 shown in Fig. 1. The development of the research aircraft, aerodynamic model, and flight test experimental implementation is described further in [8–10].

After preliminary flight testing, we discovered that the azimuth of the path initially tracked by the aircraft (before buildup of the error integral) was consistently off from the desired path by approximately 37° . PX4's extended Kalman filter (EKF) velocity estimates are obtained 100 ms after sensor readings, which, under the nominal spin rate of $375^\circ/\text{s}$, corresponds to a 37.5° delay in heading. This observation justifies a phase shift of the yaw-periodic LQ control command by this amount to account for velocity estimate delays.

For each spin maneuver, the pilot progressed through the following series of scripted test inputs:

- 1) Initiate benign, wings-level, power-off stall (δ_e : 70% trailing-edge up gradually; δ_a, δ_r : as required to level wings).
- 2) At stall break, initiate spin inputs (δ_r : 70% left; δ_e : maintain 70% trailing-edge up; δ_a : 70% left).
- 3) After two spin rotations, engage spin path control law, releasing piloted inputs.

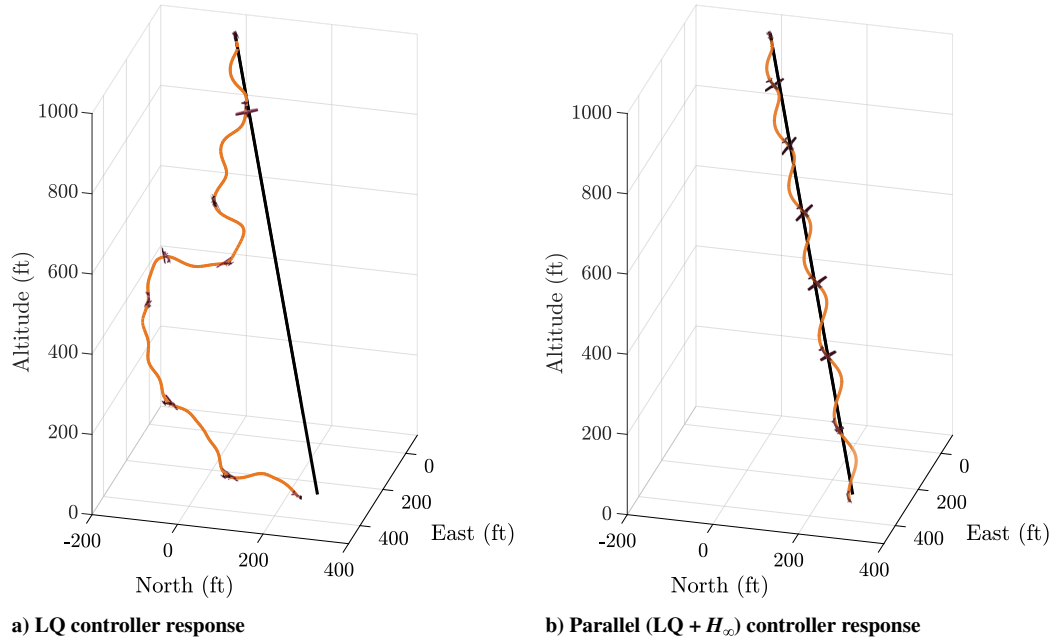


Fig. 6 LQ vs parallel (LQ + H_∞) controller simulated with wind and model perturbations.

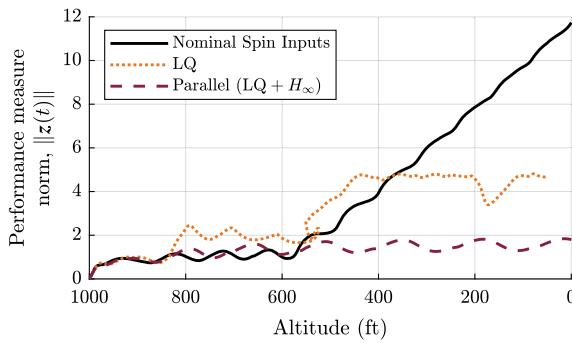


Fig. 7 Performance measure comparison.

4) At predetermined safe recovery altitude, terminate the spin path control inputs, and recover to normal flight.

The precise control surface deflections described above during spin initiation were commanded through modification of the pilot

commanded input limits, thus allowing the pilot to precisely maintain nominal spin inputs. Note that, once the control law is engaged, the piloted control inputs are no longer passed through to the aircraft. Additionally, two full rotations were allowed before control law engagement to ensure that the spin has reached its vertical, nominal motion. Each spin maneuver used for analysis included only the data once the spin path control law was engaged after development of a steady spin.

Flight test results in calm wind conditions indicate successful spin path control using the control law presented above. In Fig. 8, the desired path is shown as the solid black line, and the trajectory from flight data is shown in the orange line. As seen in Fig. 8a, the control law is capable of tracking at least a 10 deg inclination. This performance is quantified by computing the velocity vector direction error integral e_v and the non-dimensional H_∞ performance measure z from flight data using Eqs. (15) and (22), respectively. For a desired 10 deg inclination, the flight test results for e_v and z are shown in Figs. 9a and 9b, respectively. In Fig. 9b, the performance measure is broken down into the reduced state perturbation components $z_{\delta x_r}$ and computed H_∞ control components $z_{\delta u_\infty}$ to show their respective contributions. In agreement with the simulation

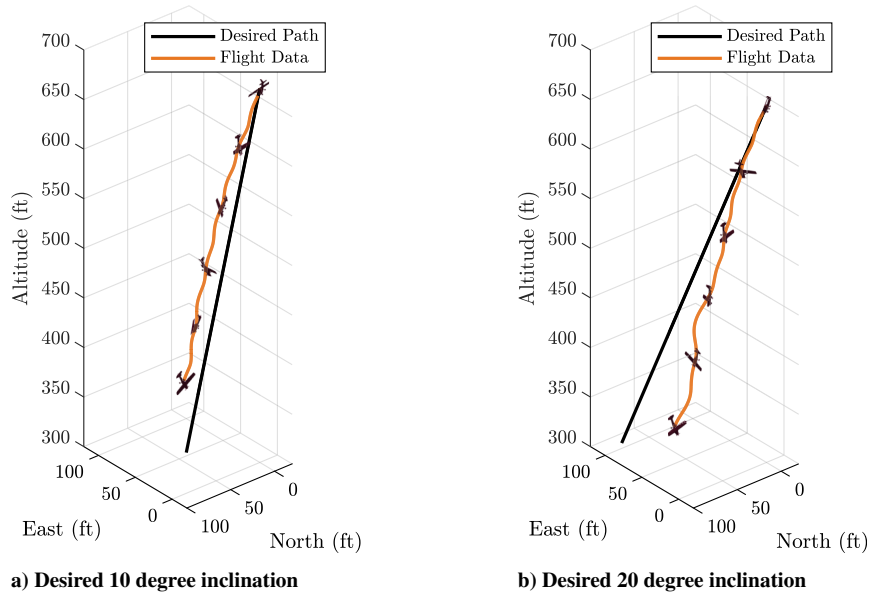


Fig. 8 Flight test results in quiescent conditions.

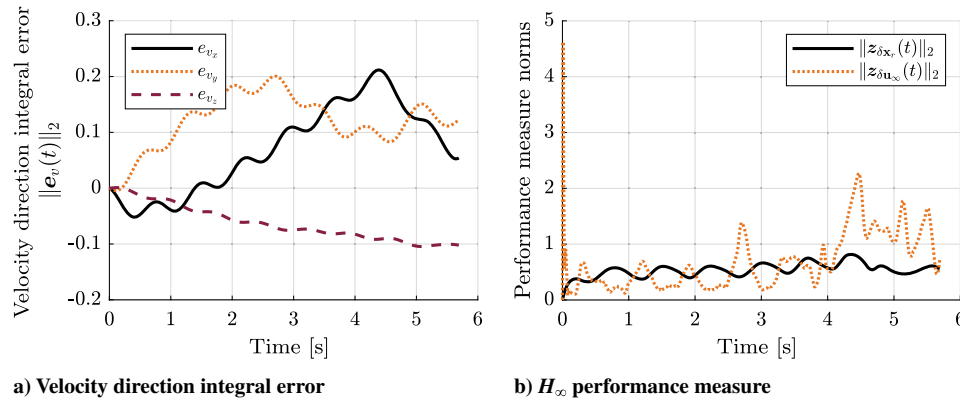


Fig. 9 Flight test performance for a desired 10 deg inclination.

results, the H_∞ performance measure remains bounded. Furthermore, Fig. 9a indicates stability of the velocity direction error integral. Recall that the control law is derived using a nominal flight, nonlinear model linearized about a spin. Even though the stall spin is not in the validated domain of this aerodynamic model, the robust H_∞ controller is still able to maintain a spin as the LQ controller drives the aircraft toward the desired path. These flight test results demonstrate that Assumption 1 is not a bad assumption for a desired 10 deg inclination.

While higher angles were tracked in simulation, the aircraft was unable to track a 20 deg inclination, at least within the limited altitude range of the test as seen in Fig. 8b. This is most likely a result of deviating too far from the aerodynamics of the nonlinear glide aerodynamic model, violating Assumption 1. Because the control law is engaged once the aircraft is in a nominal vertical spin, there is an initial deviation from the desired path due to the aircraft's momentum. The time constant associated with the convergence to this path depends on the desired inclination angle. Thus, with a higher initial altitude, it remains a possibility that tracking such an angle is achievable.

V. Conclusions

The development, simulation, and flight test of a robust, linear control law to direct the spinning descent of an aircraft along a desired azimuth and inclination angle is presented. Through an independent variable change, the yaw-periodic LQ controller stabilizes the desired velocity vector direction, accomplishing the primary objective of the control law. Then, the robust H_∞ controller guarantees (in a linear sense) a robust bound on the worst-case map from LQ controls and disturbances to perturbation from the nominal spin.

In simulations of the nonlinear dynamics, the control law shows good disturbance rejection as well as robustness to changes in the aerodynamic model. This control law is also implemented in flight test as a fixed-wing unmanned aircraft system flight termination sequence, displaying the ability to slant the spin axis up to at least a 10 deg inclination in flight with no drift in azimuth tracking. The flight test results for a 10 deg inclination show boundedness of the H_∞ performance measure z displaying the control law's robust performance in flight test where the true aerodynamics are uncertain.

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